Empirical Methods in Finance

When Bonds Stop Hedging: Evidence on Post-COVID Changes in Financial Market Risk

Research question: Has the structure of financial market risk changed since COVID-19?

Report 2

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Abstract

This project questions whether since the COVID-19 pandemic the financial market risk's structure has changed. From January 2005 to April 2026, the report focuses on a multi-asset panel composed of U.S. and European equities, emerging markets, high yield and investment grade bonds, gold, oil futures and U.S and German 10-year government bond. A risk structure is defined through three cardinal dimensions: volatility persistence, cross-asset correlation dynamics and tail risk behavior. To empirically answer the research question, the analysis applies a Dynamic Conditional Correlation filter, estimates Student-t GARCH models, assesses structural breaks using stock-bond correlations using HAC standard errors and conclude with an evaluation of portfolio implications via a 60/40 allocation benchmark and an Equal Risk Contribution strategy. The results display that in the post-COVID period, permanent increase does not come from individual asset volatility. Persistence from GARCH model stays high across considered assets, but for most series it shows a slight drop in the post-COVID period. Tail risk continues to be important, as Student-t degrees of freedom stay low. This confirms the persistence of financial returns characterized by fat-tail distribution. The project's main result is instead a breakdown on traditional stock-bond diversification. The DCC correlation shifts from a negative pre-COVID average to basically zero for the post-COVID period for the S&P 500 and the U.S. 10-year bond proxy. Furthermore, break tests seem to suggest that a stronger structural break is represented by the 2022 inflation and rate-hiking cycle. Economically speaking, these findings are significant for portfolio construction. While the classic 60/40 allocation strategy shows modest changes in performance, the base diversification benefit that comes with this strategy has weakened. After COVID, the ERC shows a degradation in performance, meaning that volatility-based allocation suffers when correlations shift. Overall, the empirical evidence presented in this report suggests that after 2020, the risk regime is characterized by asset co-movements changes and not by a more simple volatility's increase.

Keywords: COVID-19; Financial risk structure; Dynamic conditional correlation; Stock-bond correlation; Equal Risk Contribution

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1 Introduction

1.1 Economic Motivation and Background

Financial markets are historically characterized by cyclical fluctuations, where periods of low volatility and growth are interrupted by corrections and recessions. However, the COVID-19 pandemic in early 2020 has been an exogenous macroeconomic shock of a different scale. Entire sectors of the global economy were forced to shut down overnight, causing the fastest equity market crash of modern history. Observing a huge, temporary increase in volatility during a global lockdown was expected, but not every crisis implies permanent shifts in market dynamics. Therefore, the real empirical challenge, and the core focus of this project, is understanding whether the structure of risk has permanently changed in the post-pandemic world.

The years following the pandemic have been characterized by a rapid change of the macroeconomic landscape. Of central importance is the transition from a decade of ultra-low inflation and near-zero interest rates, to a regime of higher inflation, supply chain disruptions and geopolitical tensions. This changing environment has forced central banks to reconsider their positions: from strongly supporting economic expansion to aggressively fighting inflation, rapidly increasing the cost of capital and the pricing of risk for all asset classes.

This transition is important from an economical perspective. Indeed, the co-movement between stocks and bonds cannot be considered as a static parameter over the whole economic history, but is affected by factors such as market liquidit, risk premiums and and macroeconomic state (Baele et al., 2010). In times in which inflation, rather than economic growth, is the primary market concern, some risk management adjustments are required. For example the traditional negative stock-bond correlation tends to weaken, potentially requiring a deeper stress test for the classic 60/40 portfolio that relies on this relationship.

Furthermore, the 2020 shock had an unique scale in modern history, and it may have caused a "long-memory" effect in market volatility. In the pre-pandemic era high volatility regimes were often short-lived and followed by fast mean reversion. However, in the post-COVID environment, shocks appear to have higher persistence in the market, suggesting that "risk clusters" might now be more difficult to revert. To adequately capture this shift it is necessary to use dynamic econometric frameworks, such as the Dynamic Conditional Correlation (DCC) models to capture the time-varying dynamics of correlation (Engle, 2002). It has been demonstrated that international equity markets tend to correlate more strongly during crashes (Longin & Solnik, 2001), following this idea it is also interesting to understand whether this "asymmetry of correlations" has become a permanent feature of the post-pandemic regime, or just a temporary change. Indeed, if the probability of major assets crashing together has increased, the idea of "safe havens" must be re-evaluated. Understanding these changes is extremely important for solving asset allocation problems and for ensuring the stability of portfolios in a world that has structurally changed.

1.2 The Research Question

The main research question the report tries to answer is: *Can we say that, since COVID-19, the structure of risk in financial markets has changed?*

To perform a structured empirical analysis, the broader question is decomposed into three more specific, testable sub-questions:

1. Did individual asset volatility become more persistent in the post-pandemic regime?
2. Did the traditional stock-bond diversification relationship experience a structural break?
3. Did tail risk and the probability of being trapped in a "high-stress" correlation regime increase?

1.3 Defining the "Structure of Risk"

Obviously, to answer these questions with econometric rigor, it is necessary to go more in depth than simple descriptive statistics. Stating that "risk in the markets increased" just because the unconditional variance of the S&P 500 increased in March 2020 is insufficient. In this empirical project, the structure of risk is evaluated dynamically, and broke down into three distinct, quantitative dimensions:

- **Volatility Persistence and Clustering:** Risk is not constant, but it clusters over time. The first dimension analyzes whether a shock in the market decays at the same speed as it did before COVID. Indeed, if conditional volatility has become more persistent, it translates on average on longer periods of market stress when shocks happen.
- **Cross-Asset Correlation Dynamics:** Modern portfolio theory strongly relies on the diversification benefits between different asset classes. For this reason the stability of these co-movements is tested. In particular, the negative correlation between equities and government bonds has been central for asset allocation for over two decades, a structural break in this relationship would show the need for a change in asset management allocations.
- **Tail Risk and Extreme Events:** To define systemic risk it is important to analyze the behavior in the tails of the return distribution. Finally, it is analyzed whether extreme negative events have become more frequent and if the probability of major assets crashing simultaneously has increased compared to the pre-COVID baseline.

1.4 Report Structure

The report is structured as follows. The next section explains the financial and empirical literature used as a baseline to perform the analysis. Section 3 introduces the dataset, explains the necessary data transformations and and highlights the initial descriptive analysis and the unconditional market shifts between the pre- and post-COVID eras. Then the formal methodology is carefully explained, detailing the univariate GARCH specifications, the multivariate Dynamic Conditional Correlation (DCC) framework, and the Markov-switching model. Section 5 presents the main empirical results, focusing on structural break tests and shifting volatility dynamics, translating these findings into real economic results through the construction and backtesting of an Equal Risk Contribution (ERC) portfolio. To conclude, the report summarizes the main findings and discusses the limitations of empirical choices used for this report.

2 Literature Review

Analyzing the structure of risk and whether there has been a structural break has been, over the years, the subject of different financial econometric studies. To understand whether after the COVID-19 pandemic the structure of risk has changed, we must analyze the evolution of three main different but related dimensions: the persistence of market shocks, the dynamics of asset correlations over time, and the behavior of extreme tail events.

2.1 Volatility Persistence and the GARCH Framework

The first important dynamic to acknowledge, is that financial returns do not exhibit constant variance. But they actually present "volatility clustering": large shocks tend to be followed by other large shocks of either sign, and small shocks by small shocks (Mandelbrot, 1963). This suggests that the risk of an asset has at least some level of persistence. A simple moving average of past volatility is insufficient to correctly capture these dynamics. In fact, shocks in the market do not decay in a linear way, but at an exponential rate that requires a formal autoregressive structure (Bollerslev, 1986; Engle, 1982).

Bollerslev's idea of the Generalized Autoregressive Conditional Heteroskedasticity (GARCH) model is fundamental for this part of the study. Allowing the conditional variance to depend on its own past values and on its squared past returns allows to test the first dimension of our research question: the persistence of risk. This model is used to evaluate and compare the parameter measuring persistence before and after 2020. A statistically significant increase of this parameter would imply that markets have become slower to absorb shocks in the post-COVID era, leading to longer periods of distress on average.

2.2 The Dynamics of Cross-Asset Correlations

Univariate models carefully explain individual asset risk, but modern portfolio theory is based on how assets move together. Correlations have been often treated as static or estimated through rolling windows, failing to capture rapid shifts and regime breaks during crisis. However, the co-movement between different asset classes is highly unstable and reacts almost instantaneously to market news; for this reason it is often considered better to move from a static correlation matrix to models that allow the time-varying aspect of the covariance structure (Engle, 2002).

The Dynamic Conditional Correlation (DCC) model introduced by Engle (2002) has been the solution adopted in this report. In this model the covariance matrix is decomposed into individual GARCH processes and a time-varying correlation matrix, allowing to estimate large systems with multiple assets with statistical rigor. This is central in this study: the DCC model allows to exactly understand the moment when the "diversification benefit" offered by multi-asset portfolios started to decrease. The use of simpler approaches would risk to lose the breaks of 2020 and 2022 in the noise of long-term averages.

2.3 Macroeconomic Drivers and the Stock-Bond Relationship

For over two decades sovereign bonds have been used as a reliable hedge against equity drawdowns. However, this relationship is not to be given per true, but it depends on the macroeconomic environment. Indeed, the sign and magnitude of the stock-bond correlation are mainly driven by the volatility of inflation and monetary policies: usually when there is more concern around inflation than economic growth, fixed income securities tend to lose some of their hedging properties (Campbell et al., 2017).

Therefore, the fast post-COVID transition from a low-inflation environment to a period of financial distress and high interest rates may represent a potential point of structural break in traditional portfolio allocations, such as the 60/40 strategy. Testing for structural breaks in the

dynamic correlation series, it is empirically evaluated if this major shift in the macroeconomic environment has altered the relationship between asset classes, in particular between equities and sovereign bonds.

2.4 Tail Risk and Correlation Asymmetries

Often it is assumed that correlations in normal times are the same as correlations during crashes. However, empirical evidence demonstrates that the diversification effect that investors rely on, often vanishes in periods of severe market stress. Assets tend to correlate stronger during large crashes than during upward trends; in fact, this "correlation asymmetry" means that systemic risk is usually underestimated by standard linear models (Longin & Solnik, 2001).

To address this risk, it is followed the methodology proposed for heteroskedasticity time series, combining GARCH modeling with a focus on the distribution's tails. Literature shows that residuals of financial models often exhibit "fatter tails" than normal gaussian distributions, and for this reason to estimate risk measures like Value-at-Risk, one must specifically account for the higher probability of extreme negative events for example (McNeil & Frey, 2000).

2.5 Regime Shifts and Markov-Switching Models

Finally, it is rare that the transition between different market environments is a smooth and linear process. Instead, markets usually move across distinct "regimes", such as a low-volatility/low-correlation state and a high-volatility/high-correlation state. It has been empirically demonstrated that these regime shifts can be modeled as a process where the probability of being in a certain state depends on the previous period's state (Hamilton, 1989).

The Markov-switching framework developed by Hamilton (1989) is used as a basis in this report' analysis. The model is used to determine if the post-COVID era is simply a temporary period of stress or if the market now finds itself in a high-risk regime with a high probability of staying there. Calculating the transition probabilities between these states it is possible to understand in a statistical and probabilistic way if the pandemic caused a permanent shift into a more unstable market regime.

3 Data

3.1 Variables

This report is based on a multi asset data panel that includes the main sources of financial risk across equity markets, fixed income, commodities, credit and interest rates. The dataset spans from 31 December 1989 to 24 April 2026 and it contains 14 financial series. This analysis keeps only nine of these fourteen assets, as they represent the main risk factors that are needed to study whether COVID-19 changed the financial structure of risk.

The selected variables are the S&P 500, Euro Stoxx 50 and MSCI Emerging Markets indices for equity risk; US investment grade and US high yield bonds for fixed-income and credit risk; gold and oil futures for commodity and inflation-related risk; and the US and German 10-year government yields for interest-rate risk. This analysis does not include the Hang Seng and the SMI as they may be redundant from a risk perspective as they are highly correlated with the broader equity indices, where diversification depends on cross asset correlations (Goetzmann et al., 2005). FX series are also not taken into account for this report as they are known to be difficult to forecast, especially out of sample where they are close to a random walk benchmark (Meese & Rogoff, 1983).

The considered sample begins in January 2005. This methodological choice allows to avoid the early part of this dataset while preserving a long enough period before the COVID crisis. Once all variables are selected, the analysis begins on resampling the data to a business day calendar and forward filled. The final version of the usable dataset has 5'560 daily observations with no missing values, with the considered time window spanning from 3 January 2005 to 24 April 2026.

To analyse the effect of COVID-19 on the risk structure, this sample is divided into three distinct periods. The pre-COVID period stands as the baseline regime and it starts on 4 January 2005 and runs until 31 December 2019. The year 2020 acts as a buffer year. The post-COVID time period starts on 1 January 2021 and ends with the last available sample's date, the 24 April 2026. This time division allows the report to better compare the pre-COVID financial risk structure to the more recent financial environment with more inflation and rate hiking events, without mixing the transition year into either regime.

3.2 Data Transformations

For price and index series, returns are transformed in continuous and compounded log returns:

$$r_{i,t} = \ln\left(\frac{P_{i,t}}{P_{i,t-1}}\right), \tag{1}$$

where $P_{i,t}$ is the price of asset i at time t, and $r_{i,t}$ is its log return. This transformation is applied to the S&P 500, Euro Stoxx 50, MSCI EM, US IG bonds, US HY bonds, gold and oil futures.

For the yield variables, the analysis computes daily changes in the 10-year yields:

$$\Delta y_{i,t} = y_{i,t} - y_{i,t-1}. \tag{2}$$

GARCH, DCC and portfolio models in this report are computed based on return series rather than yield levels. For this reason the yield changes $\Delta y_{i,t}$ are then converted into a bond return proxies with a standard first order duration formula:

$$r_{i,t}^{bond} \approx -D_i \cdot \frac{\Delta y_{i,t}}{100}. \tag{3}$$

The negative sign represents the inverse relation between yield and bond prices. This linearization is commonly used to approximate a bond portfolio's price sensitivity to interest rate changes

(Fisher & Weil, 1971; Ingersoll et al., 1978). In this report, the empirical implementation used for the modified duration D_i is set for 8.5 years for the 10 year US bond, while for the german 10 year yield is set at 9.0 year.

Finally, the analysis computes a risk free proxy from the 10 year US yield level, used for example to calculate the Sharpe Ratio:

$$r_t^{f,daily} = \frac{\ln\left(1 + y_t^{US10Y}/100\right)}{252}, \tag{4}$$

$$r_t^{f,weekly} = \frac{\ln\left(1 + y_t^{US10Y}/100\right)}{52}. \tag{5}$$

It is important to say that this is not a perfect approximation of the risk free level for the time considered since no 3-month Treasury bill is available. However, this implementation is transparent as it follows official guidelines and uses only information that are available in the provided dataset.

3.3 Descriptive Statistics, Autocorrelation and Cross-Asset Dynamics

Table 1: Summary Statistics for Daily Transformed Variables

Variable	Mean	Ann. Vol.	Skewness	Exc. Kurtosis	JB stat.
S&P 500	0.0003	0.1878	-0.4795	13.7936	44283.0
Euro Stoxx 50	0.0001	0.2069	-0.3124	8.4369	16577.9
MSCI EM	0.0002	0.1871	-0.5309	8.4616	16845.3
US IG Bonds	0.0002	0.0520	-0.6768	7.1792	12362.7
US HY Bonds	0.0002	0.0512	-1.8092	32.1967	243141.7
Gold	0.0004	0.1756	-0.4430	6.4673	9869.7
Oil Futures	0.0003	0.4118	0.0115	18.5431	79615.1
US 10Y Change	0.0000	0.9034	-0.1318	3.0034	2105.4
German 10Y Change	-0.0001	0.7104	-0.0079	2.7275	1723.2

Note: The table reports daily means and annualized volatility. Jarque-Bera p-values are equal to 0.0000 for all series, so normality is rejected at the 1% level.

The principal statistics for daily transformed series are reported on Table 1. This table shows that the series are non-Gaussian. All assets are not normal under the null hypothesis as they fail the Jarque Bera normality test at 1% significance level. Equity return show a general negative skewness, with US high yield bonds carrying the most extreme left tail behaviour, with $\hat{S} = -1.8092$ and the largest excess kurtosis, with $\hat{K} - 3 = 32.1967$. From a price and index return perspective, oil futures are the class showing the highest volatility, with an annualized value equal to $\hat{\sigma}_{ann} = 41.18\%$. The US and german 10 year yield actually display a higher annualized volatility, but these results should not be interpreted the same way as the asset-return volatility. This is because the report used a yield-change units rather than a return units as for the others asset classes. Overall, this brief summary confirms the presence of extreme observations and fat tails, which supports the use of special volatility models that include non-Gaussian innovations.

The report also computes the ARCH-LM and Ljung-Box tests to squared returns, which exclude the null hypothesis stating the presence of volatility clustering across asset classes. In other words, large shocks tend to be followed by other large shocks while calm periods tend to stay so. The methodology follows this empirical property as it motivates the implementation of a GARCH model rather than a static and constant volatility model.

4 Methodology

4.1 Empirical Strategy

The project's empirical strategy seeks to assess whether the risk structure in financial market has changed since the advent of the COVID-19 pandemic. Risk's structure is studied along four dimensions. First, the study examines whether the volatility became more persistent after COVID. Second, it studies if the cross-asset correlations changed. Third, the analysis checks whether the post- COVID time window can be considered as a structural break or only as a different risk regime. Finally, it converts these results in a 60/40 portfolio and an equal risk contribution portfolio. The analysis starts with descriptive statistics, then computes normality test and volatility-clustering diagnostics. The report then tries to estimate GARCH models in order to model conditional volatility. Once the volatility is filtered out, residuals are then used in a Dynamic Conditional Correlation model (DCC). Pre-COVID and post-COVID periods are analyzed using a series of instruments such as rolling correlations, DCC correlation, Welch mean comparison and formal regression to evaluate a possible structural break in financial markets. This produces both statistical and visual evidence on how and if the diversification pattern changed after the COVID- 19 crisis. The report conducts a two state Markov switching model that is applied to the selected stock bond correlation series. The portfolio section of this report evaluates whether this change in correlation matters for investors.

4.2 Model Specification

This model utilizes several empirical models. The first one is the GARCH model, where for each asset i, returns are computed as:

$$r_{i,t} = \mu_i + \varepsilon_{i,t} \qquad (6) \qquad\qquad \varepsilon_{i,t} = \sigma_{i,t} z_{i,t} \qquad (7)$$

where $z_{i,t}$ follows a Student-t distribution. the conditional variance follows:

$$\sigma^2_{i,t} = \omega_i + \sum_{j=1}^{p} \alpha_{ij} \varepsilon^2_{i,t-j} + \sum_{k=1}^{q} \beta_{ik} \sigma^2_{i,t-k}. \qquad (8)$$

In the GARCH model, volatility's persistence is measured with the value obtained by $\sum \alpha + \sum \beta$. For this report, the use of Student-t innovations is supported by the observed fat tails in the distribution.

Then, the analysis computes the standardized residuals as follow:

$$z_{i,t} = \frac{\varepsilon_{i,t}}{\sigma_{i,t}}. \qquad (9)$$

These residuals serve as inputs for the DCC filter. The formula used for the DCC quasi-correlation matrix is described as

$$Q_t = (1 - \theta_1 - \theta_2)\bar{Q} + \theta_1 z_{t-1} z'_{t-1} + \theta_2 Q_{t-1}, \qquad (10)$$

Rather than estimating the DCC parameters, in this implementation these values are calibrated. The values $\theta_1 = 0.05$ and $\theta_2 = 0.93$ are calibrated in order to produce a highly persistent correlation process. Since $\theta_1 + \theta_2 = 0.98 < 1$, DCC stability conditions is satisfied.

In this report, DCC filter are not used to estimates structural DCC parameters but rather to obtain a stable measure of the time variation across asset correlations.

By rescaling the following:

$$R_t = \text{diag}(Q_t)^{-1/2} Q_t \text{diag}(Q_t)^{-1/2}. \qquad (11)$$

the report obtain the conditional covariance matrix as:

$$H_t = D_t R_t D_t, \tag{12}$$

where D_t represents the diagonal matrix of GARCH conditional volatilities. The DCC model is computed based on daily standardized residuals, while the weekly returns are used when rolling-correlation checks are computed to reduce short-term noise.

To answer the report's question, the analysis then focus itself on the Welch t-test applied to the pre-COVID and post-COVID average correlations. It also estimates a structural break regression for the main stock-bond DCC series as:

$$\rho_t = \alpha + \beta \mathbf{1}_{t \geq t^*} + u_t. \tag{13}$$

The β coefficient shows the change in average correlation for the period after the candidate break regime date. The analysis uses Newey and West (1987) HAC standard errors to track autocorrelation in the correlation series. This project considers two dates: March 2020, when COVID broke out, and January 2022 representing a moment of general inflation and rate hiking shock.

a two state Markov-switching model is computed on the weekly stock-bond DCC correlation:

$$\rho_t = \mu_{s_t} + \varepsilon_t, \qquad s_t \in \{1, 2\}. \tag{14}$$

This model allows mean and variance to diverge across different regimes.

The project then computes two different portfolio specifications. The first one is a simple 60/40 stock-bond allocation:

$$r_t^{60/40} = 0.60 r_t^{SP500} + 0.40 r_t^{US10Ybond}. \tag{15}$$

The second portfolio is the equal risk contribution portfolio. The weights are obtained as:

$$\min_w \sum_{i=1}^{N} \left(\frac{w_i (H_t w)_i}{w' H_t w} - \frac{1}{N} \right)^2 \quad \text{s.t.} \quad \sum_i w_i = 1, \quad w_i \geq 0. \tag{16}$$

4.3 Estimation Procedure

The project estimates all model on aligned daily return samples, spanning from 4 january 2005 to 24 april 2026. Pre-COVID, COVID and post-COVID periods respectively contain 3'911, 262 and 1'386 daily observations. The analysis excludes the year 2020 for the formal pre/post comparison as it treats this year as a transition period.

For volatility models, the study considered four different specifications for every asset: ARCH(1), ARCH(2), GARCH(1,1) and GARCH(2,1). To determines which specification is the best, models are then ranked using the Bayesian Information Criteria (BIC):

$$BIC = k \ln(n) - 2\ell, \tag{17}$$

where k is the number of parameters, n is the number of observations and ℓ is the maximized log-likelihood. Student-t innovations tests then help select the model on the full sample. On top of that, an additional common GARCH(1,1) specification is estimated for both pre and post-COVID periods. This specification is used so that persistence's changes are motivated by data rather than by model order changing.

DCC filter is then used on the full set of GARCH residuals. The analysis mainly focuses on six pairs of correlations series, each with the S&P 500. These pairs are S&P 500 with US IG bonds, the US 10-year bond proxy, gold, oil, US high yield bonds and Euro Stoxx 50. For each couple, pre and post-COVID mean correlations are set side by side using a Welch t-tests

computation. The principal stock-bond pair is also measured with the HAC structural break regression model discusses in above in this report.

To be more empirically robust, the report computes an additional robustness check. Using a EWMA filter with $\lambda = 0.94$ dynamic correlations are also computed. This method is useful as it does not directly depend on the GARCH-DCC structure. The analysis adds a dummy regression with HAC standard errors applied to the EWMA correlations cited before in order to verify whether the main project's conclusion is sensitive to the chosen correlation model.

Finally, portfolio implications are evaluated through a series of parameters, such as annualized return and volatility, Sharpe ratio and maximum drawdown. The computed 60/40 portfolio allocation come directly from S&P 500 returns and the 10 year US bond proxy. The ERC portfolio described before is rebalanced every month using lagged DCC covariance matrices that avoids look ahead bias. This final contribution relies the econometric and empirical evidence to the practical question that this study wants to answer.

5 Results

5.1 Main Findings

Volatility Persistence and Tail Risk

The well-known stylized facts of financial returns documented since Mandelbrot (1963) and Bollerslev (1986) are confirmed by the full-sample GARCH estimates. Persistence parameters $\hat{\alpha}+\hat{\beta}$ are close to one for all assets, this indicate that Volatility Shocks decay slowly, $\hat{\nu}$ (degrees-of-freedom parameter) remains below 10 for every asset, confirming returns have fat tails even after the GARCH structure absorbs the conditional heteroskedasticity.

Table 2: Pre/Post COVID GARCH(1,1) — Persistence and Tail Thickness

	Pre-COVID (2005–2019)			Post-COVID (2021–2026)		
Asset	$\hat{\alpha}+\hat{\beta}$	$\hat{\nu}$	n	$\hat{\alpha}+\hat{\beta}$	$\hat{\nu}$	n
S&P 500	1.000	4.53	3,911	0.981	5.81	1,386
Euro Stoxx 50	1.000	5.48	3,911	0.930	5.46	1,386
MSCI EM	0.986	8.75	3,911	0.962	6.79	1,386
US IG Bonds	0.995	10.69	3,911	0.991	7.84	1,386
US HY Bonds	1.000	5.08	3,911	0.974	4.47	1,386
Gold	0.997	4.98	3,911	0.984	5.13	1,386
Oil	1.000	6.62	3,911	0.991	6.04	1,386
US 10Y proxy	0.996	11.00	3,911	0.989	8.40	1,386
DE 10Y proxy	0.995	8.79	3,911	0.991	7.62	1,386

The post-COVID persistence is marginally lower for every asset, this suggest that volatility decay slightly faster now than in the pre-pandemic regime. European equities had the most pronounced drop ($1.000 \rightarrow 0.930$), more evident is the change in the tail thickness parameter $\hat{\nu}$: falling for Emerging Markets ($8.75 \rightarrow 6.79$), US IG bonds ($10.69 \rightarrow 7.84$) and US Sovereign Bonds ($11.00 \rightarrow 8.40$). Fatter tails are implied by lower ν, this means that extreme events become more likely once we remove the GARCH volatility filter. The single-variable evidence suggests that volatility is not the dominant source of regime change.

The Stock–Bond Diversification Breakdown

The most important change in the risk structure of financial markets is shown by the DCC filters. The daily DCC correlation between the S&P 500 and each of six counterparts is shown in the Figure 1 with pre and post COVID averages shown as reference lines.

The stock–bond pair is the most economically significant result: the conditional correlation between the S&P 500 and the US 10-year government bond proxy averaged -0.325 pre-COVID and -0.040 post-COVID, a shift of $+0.285$ that brings the relationship close to zero, while the correlation with US investment grade credit moves even further from -0.222 to $+0.083$.

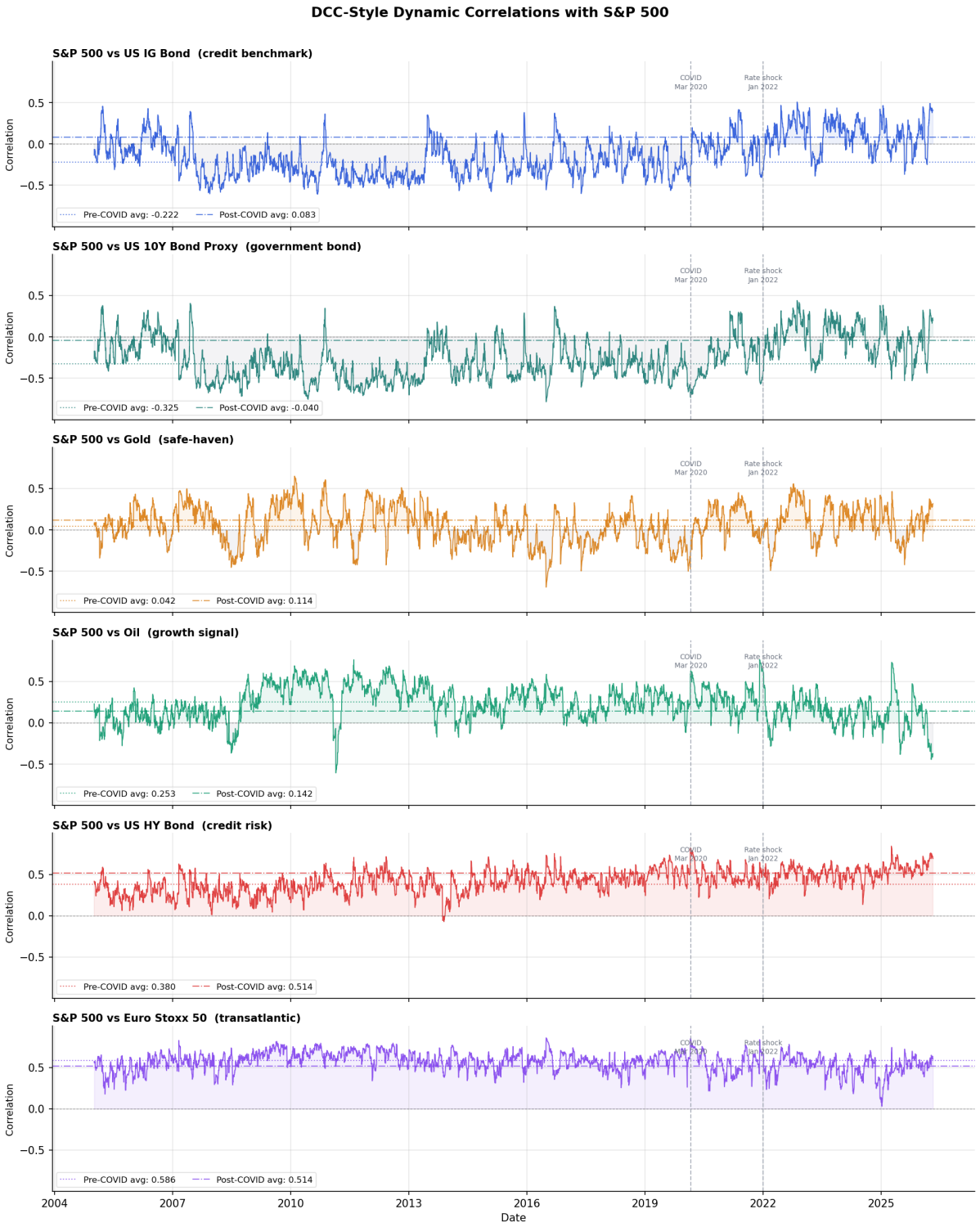


Figure 1: DCC dynamic correlations between the S&P 500 and the six counterparts (2005–2026). The dashed lines indicate pre- and post COVID averages. The top two bond panels show the shift from negative correlation to near-zero post-COVID correlations.

A Welch t-test formally rejects equality of the pre- and post-COVID mean correlation for every pair tested. Table 3 reports the results for the six pairs.

Table 3: DCC Pre vs Post COVID Welch t-test Results

Pair	$\bar{\rho}_{pre}$	$\bar{\rho}_{post}$	$\Delta\bar{\rho}$	t-statistic	p-value
S&P 500 – US 10Y proxy	−0.325	−0.040	+0.285	+41.82	<0.0001
S&P 500 – US IG Bonds	−0.222	+0.083	+0.305	+49.07	<0.0001
S&P 500 – US HY Bonds	+0.380	+0.514	+0.134	+39.14	<0.0001
S&P 500 – Gold	+0.042	+0.115	+0.073	+11.27	<0.0001
S&P 500 – Oil	+0.253	+0.142	−0.111	−17.65	<0.0001
S&P 500 – Euro Stoxx 50	+0.586	+0.514	−0.072	−19.42	<0.0001

Defensive assets that hedged equity risk before (such as government bonds, investment grade credit and gold), now move more closely together, reducing their hedging role. In contrast, oil and European equities which co-moved positively with the S&P 500, became less correlated. This show that we are in a regime in which the dominant risk factors has shifted: When asset prices are driven by inflation instead of economic growth, bonds become less useful as a hedge because rising interest rates can hurt both stocks and bonds at the same time (Baele et al., 2010).

Identifying the Break: 2020 or 2022?

Was the observed shift caused by the COVID-19 pandemic, by the the inflation and rate-hiking cycle that followed in 2022 or by both? To answer this, the following regression is estimated with Newey-West HAC standard errors for two candidate break dates t^*:

$$\rho_t = \alpha + \beta \mathbf{1}_{t \geq t^*} + u_t. \tag{13}$$

Table 4: Structural Break Tests on the Stock–Government-Bond DCC Series

Break date t^*	$\hat{\alpha}$	$\hat{\beta}$	HAC t-stat	p-value	95% CI for $\hat{\beta}$
2020-03-01	−0.327	+0.244	8.62	<0.0001	[+0.189, +0.300]
2022-01-01	−0.317	+0.297	10.52	<0.0001	[+0.242, +0.353]

Both break dates produce statistically significant shifts in the stock-bond correlation, but the 2022 coefficient (+0.297) is larger than the 2020 coefficient (+0.244), combining this result with the visual evidence in Figure 1 we can say that the 2022 rate-hiking cycle, not the pandemic itself is the dominant structural break. The COVID episode triggered a correlation spike that subsequently mean-reverted, only when inflation became persistent and rates began to rise did the negative stock–bond correlation change into a near-zero or positive relationship

A two state Markov-switching regression using the weekly stock-government-bond DCC series supports this result (Hamilton, 1989). The estimated regimes have mean correlations of −0.435 (low-correlation regime) and −0.028 (high-correlation regime). The sample spends 55.3% of weeks in the low-correlation regime and 44.7% in the high-correlation regime, with the second one found almost only in the post-2022 period (Figure 2).

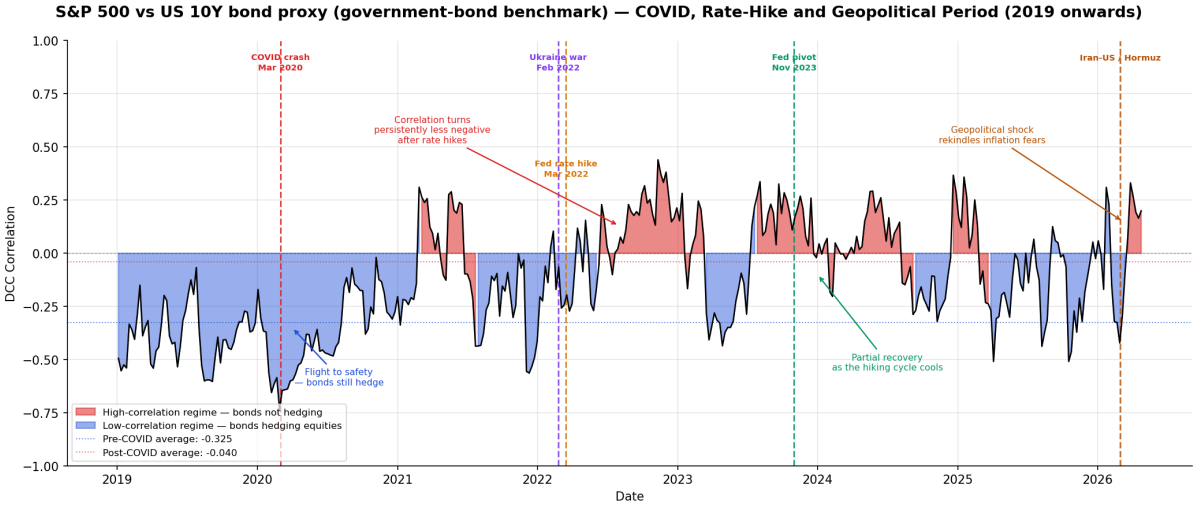


Figure 2: Stock–bond DCC correlation between the S&P 500 and the US 10Y bond proxy from 2019–2026. Red areas show periods of high correlation, blue areas show periods of low correlation where bonds still hedge equities.

5.2 Portfolio Implications

Two complementary exercises are conducted since the change in conditional correlation structure has direct consequences for portfolio construction: a static 60/40 benchmark and a fully dynamic Equal Risk Contribution (ERC) portfolio. The 60/40 portfolio holds 60% S&P 500 and 40% US 10Y government-bond return proxy, rebalanced daily. Annualised volatility increased from 9.98% pre-COVID to 10.62% post-COVID, the sharpe ratio remained stable (0.14 vs 0.15), thanks to a slightly higher average return. Maximum drawdown was smallet in the post-COVID period (-22.96% vs -35.75%), but this comparison is distorted because the pre-COVID sample includes the 2008 Global Financial Crisis. The key evidence is the cross-correlation structure of the two assets, while the overall portfolio results remain similar, the diversification benefit has largely disappeared, bonds no longer provide the negative correlation that used to protect the portfolio during equity losses.

Table 5: 60/40 Portfolio Performance — Pre vs Post COVID

	Pre-COVID	Post-COVID
Annualised Return	4.32%	4.95%
Annualised Volatility	9.98%	10.62%
Sharpe Ratio	0.14	0.15
Maximum Drawdown	-35.75%	-22.96%
Observations	3,911	1,386

The dynamic ERC portfolio confirms this. Weights are rebalanced monthly using the DCC-implied conditional covariance matrix H_t:

$$\min_{w} \sum_{i=1}^{N} \left(\frac{w_i (H_t w)_i}{w' H_t w} - \frac{1}{N} \right)^2 \quad \text{s.t.} \quad \sum_i w_i = 1, \quad w_i \geq 0. \tag{16}$$

The ERC Sharpe ratio drops from $+0.29$ pre-COVID to -0.25 post-COVID. The change is clear, Risk-parity allocation, which depends on stable cross-asset correlations to generate diversification perform worse in the post-COVID period. Figure 3 shows how the monthly ERC

weights changed over the full sample. The allocation is quite stable in the pre-COVID period for government and investment grade bonds receiving a large share of the risk budget, due to their low volatility and negative correlations with equities. The bond allocation shrinks after 2022 and the portfolio concentrates more capital in gold and in low-volatility equity as a response to the loss of diversification benefits from fixed income.

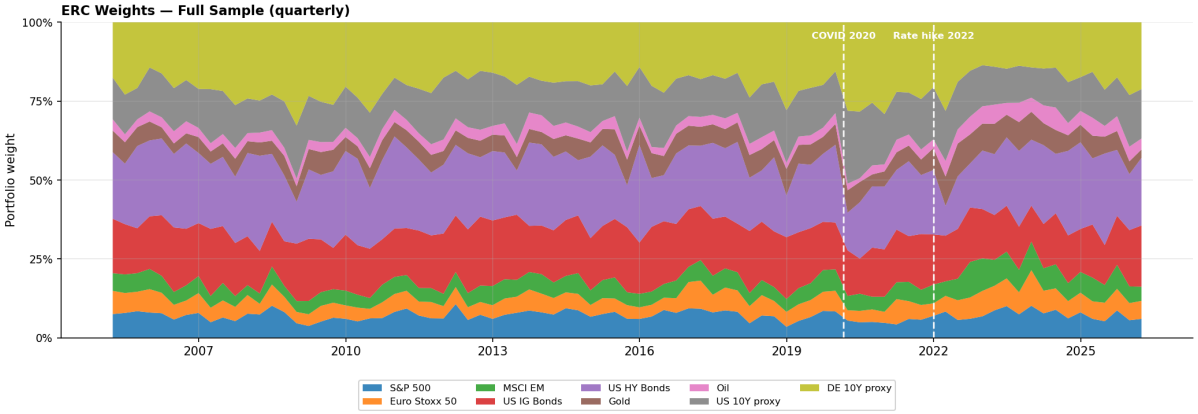


Figure 3: Monthly ERC portfolio weights (2005–2026). The graph shows how the portfolio weights changed over time across the nine assets. The two vertical lines mark the COVID crash (March 2020) and the start of the Fed rate-hiking cycle (January 2022).

Robustness — EWMA Correlations

To verify the results, the structural-break exercise is repeated using EWMA dynamic correlations with decay parameter $\lambda = 0.94$:

$$\rho_t = \frac{cov_{12,t}}{\sqrt{var_{1,t} \cdot var_{2,t}}} \qquad cov_{12,t} = (1-\lambda)\, r_{1,t-1} r_{2,t-1} + \lambda\, cov_{12,t-1} \tag{18}$$

computed on raw returns instead of GARCH-filtered residuals. The conclusion doesn't change:the stock–investment-grade-bond shift is +0.43 (HAC $p < 10^{-35}$) and the stock–high-yield shift is +0.23 (HAC $p < 10^{-33}$). The results are reinforced because both a fully parametric DCC model and a non-parametric EWMA approach produce statistically significant results at the four-decimal level, strengthening the credibility of the results.

5.3 Economic Interpretation

Volatility and tail thickness changed marginally, the most significant change is in the correlation matrix itself. The stock–bond hedge that defined two decades of asset allocation, from the early 2000s through to 2019, no longer functions in the post-2022 inflation regime. This is consistent with Baele et al. (2010) and Driessen et al. (2003), who show that the sign of the stock-bond correlation depends on the dominant macroeconomic shock: when growth risk dominates bonds hedge equities but when inflation risk dominates, both asset classes fall together.

The pandemic itself was not the source of the regime change. COVID produced a sharp but mean-reverting correlation spike; the shift came two years later, when inflation became persistent, the Fed became more hawkish and began an aggressive cycle of interest-rate hikes. This is exactly a macroeconomic regime change that Hamilton (1989) describes as a low-frequency switch in the underlying state, rather than a temporary shock.

The persistence of fat tails and the breakdown of negative correlations have direct portfolio consequences. ERC Sharpe ratio deterioration shows that risk-parity strategies are especially vulnerable to the new regime. Investors who relied on this correlation as a structural feature of markets must now treat it as a state-dependent variable.

6 Limitations

First, the post-COVID sample contains only 1,386 daily observations (approximately 5.3 years), which is short relative to the time scales at which structural macroeconomic regimes evolve. The stock-bond correlation may revert if inflation regime fades and growth shock return to dominate. The results presented cannot prove that this new regime is permanent Second, the analysis uses duration-based proxies to convert yield changes into approximate bond price returns, with modified duration of 8.5 years for the US 10Y and 9.0 years for the DE 10Y:

$$r_{i,t}^{bond} \approx -D_i \cdot \frac{\Delta y_{i,t}}{100}. \tag{3}$$

This linear approximation (Fisher & Weil, 1971; Ingersoll et al., 1978) is accurate for small yield changes but underestimate bond losses during the large yield movements of the 2022 rate-hiking cycle. As a result, the DCC and structural break results may be conservative, meaning that the true breakdown in stock-bond diversification could be larger than reported here.

Third, the DCC parameters $(\theta_1, \theta_2) = (0.05, 0.93)$ are fixed at conventional values rather than estimated jointly with the marginal GARCH parameters. A fully estimated DCC would allow the data to determine how rapidly correlations respond to new information. However, joint estimation of a nine-asset DCC introduces well-known numerical instability (Engle, 2002), and sensitivity checks with $(\theta_1, \theta_2) = (0.02, 0.97)$ yield qualitatively identical conclusions.

Fourth, the structural break tests use pre-selected break dates: March 2020 and January 2022. A data-driven method would be more robust, but these dates are economically motivated by well-known macroeconomic events. Therefore, the main interpretation of the results is unlikely to change.

Finally, ERC portfolio is constructed without transaction costs, short-selling constraints beyond non-negativity or position limits at the asset class level. The decline in the ERC Sharpe ratio should be read as a directional finding about the deterioration of the underlying diversification structure, not as a backtest of a strategy.

7 Conclusion

The project aims to understand whether the structure of market risk changed after COVID-19 pandemic. From the results it can be concluded that individual asset volatility has stabilized after the initial shock, but the structure of asset co-movements changed and going through a structural break. DCC model allowed to verify that the post-2020 market presents a different setting of how risks interact across different asset classes, rather than temporary volatility spikes.

The main evidence of this structural break is the change in the traditional stock-bond diversification mechanism. Dummy regressions and Markov-switching framework analysis showed that the negative relationship between the S&P 500 and the 10-year US Treasury bonds has decreased. The correlation moved from a baseline of -0.31 before the pandemic, to a much weaker correlation of -0.10, with the breakdown accelerating around January 2022. This change is aligned with financial literature: when concerns about inflation and monetary policies dominate the growth dimension, the stock-bond covariance becomes positive (Campbell et al., 2017).

These results carry some implications for risk management and asset allocation. The traditional 60/40 portfolio is less effective when the hedging asset moves together with stocks during shocks. The analysis on Equal Risk Contribution (ERC) strategy shows that traditional risk-budgeting models also become less reliable in this new regime. Indeed, ERC optimizes weights based on relative volatility levels, over-weighting bonds because of their lower nominal variance, ignoring the break in co-movements. Then, the ERC portfolio fails to protect capital when asset classes drop together. Concluding, modern risk management must evolve beyond simple volatility-based budgeting to handle this persistent high-inflation environment.

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A Additional Figures

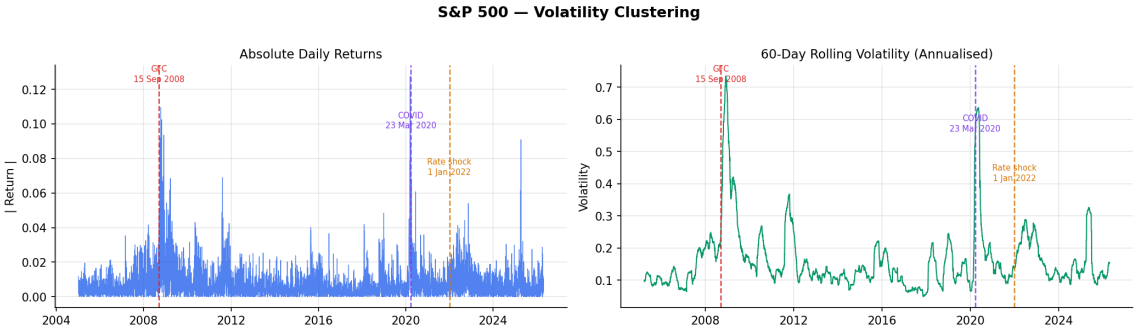


Figure 4: Volatility clustering is visible across the main asset returns, confirming time-varying risk dynamics.

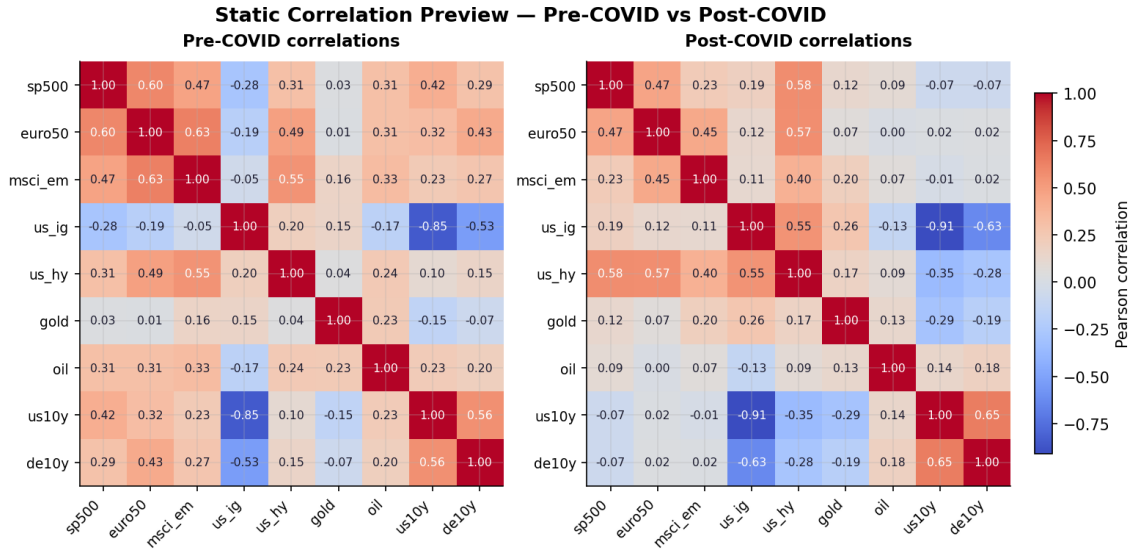


Figure 5: Static pre- and post-COVID correlation matrices show broad changes in cross-asset dependence.

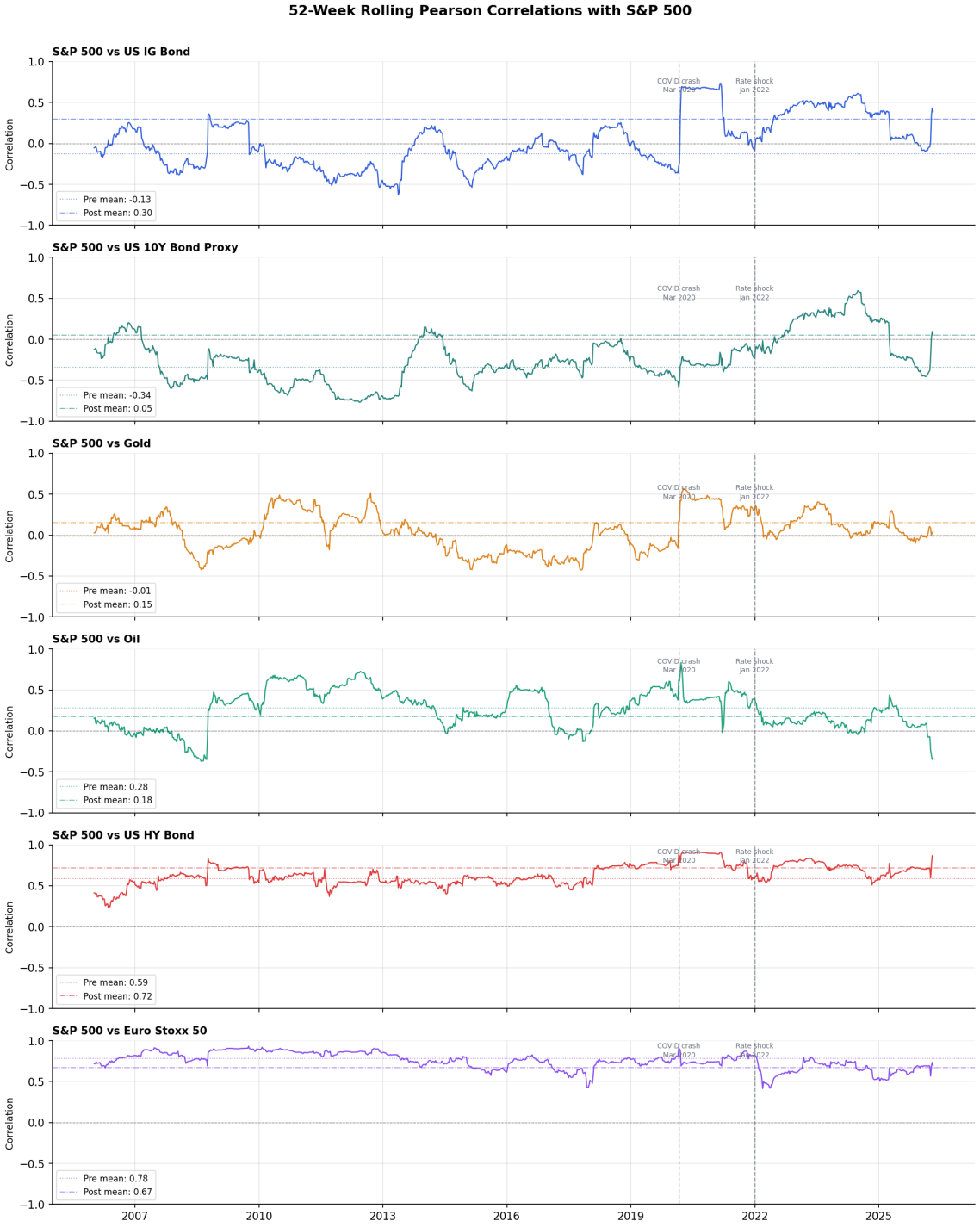


Figure 6: Rolling correlations illustrate the time variation in the relationship between the S&P 500 and selected assets.

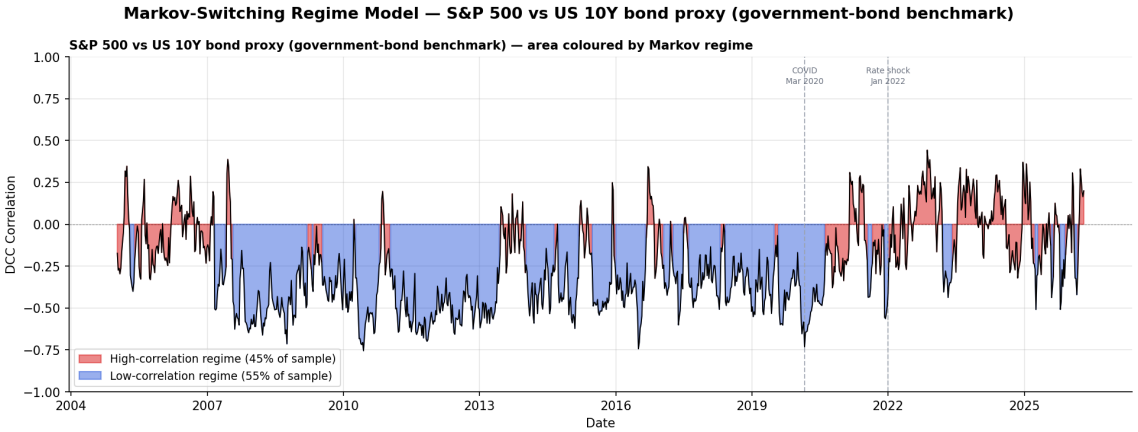


Figure 7: The Markov-switching model identifies periods of low- and high-correlation regimes in the stock-bond relationship.

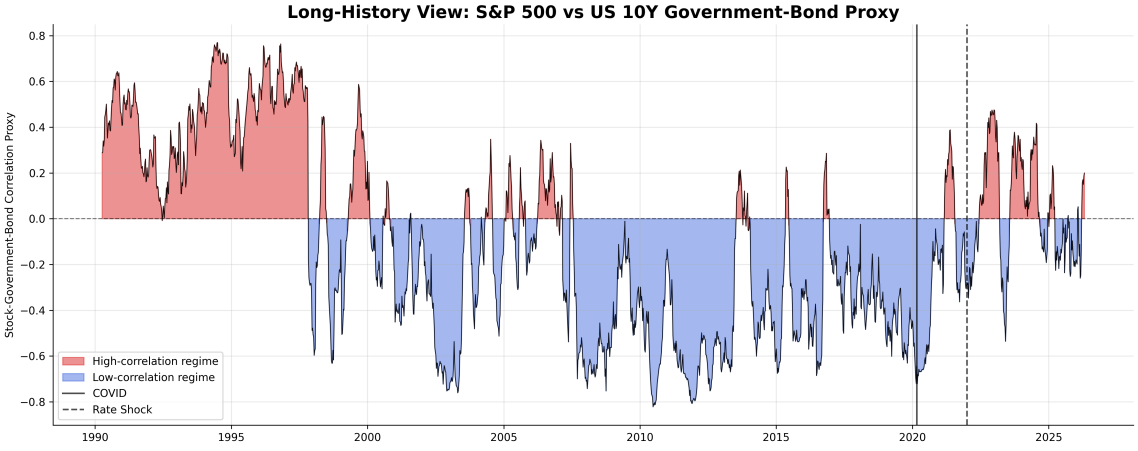


Figure 8: The long-sample regime classification highlights how unusual the post-2022 correlation environment is.

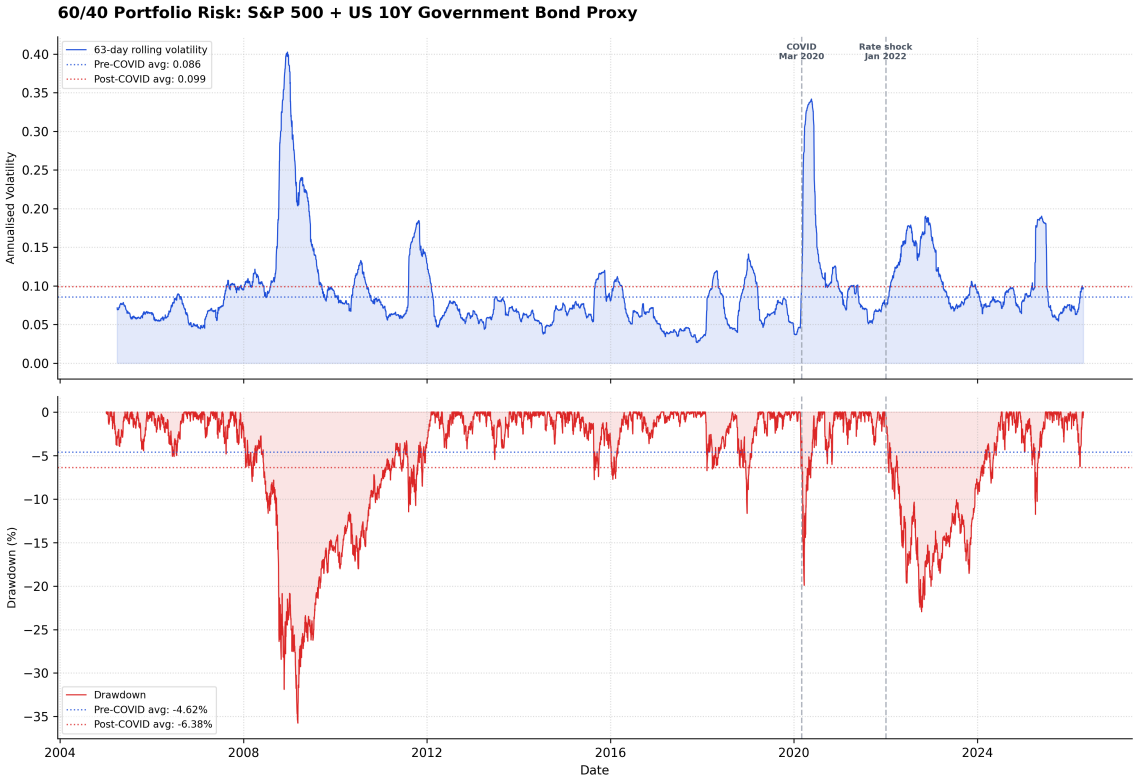


Figure 9: The 60/40 portfolio experienced only modest headline performance changes despite weaker diversification.

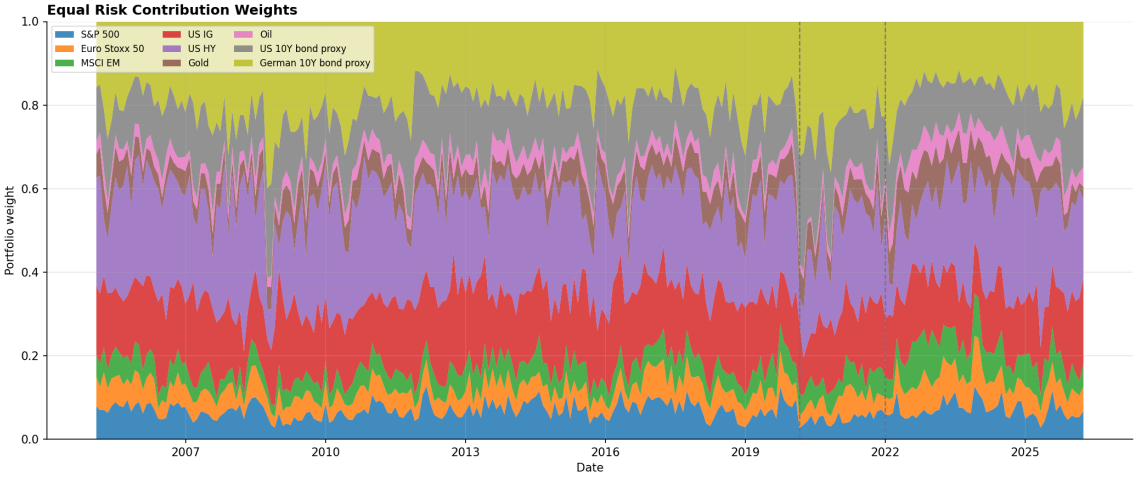


Figure 10: ERC portfolio weights evolve over time as the covariance structure changes across regimes.

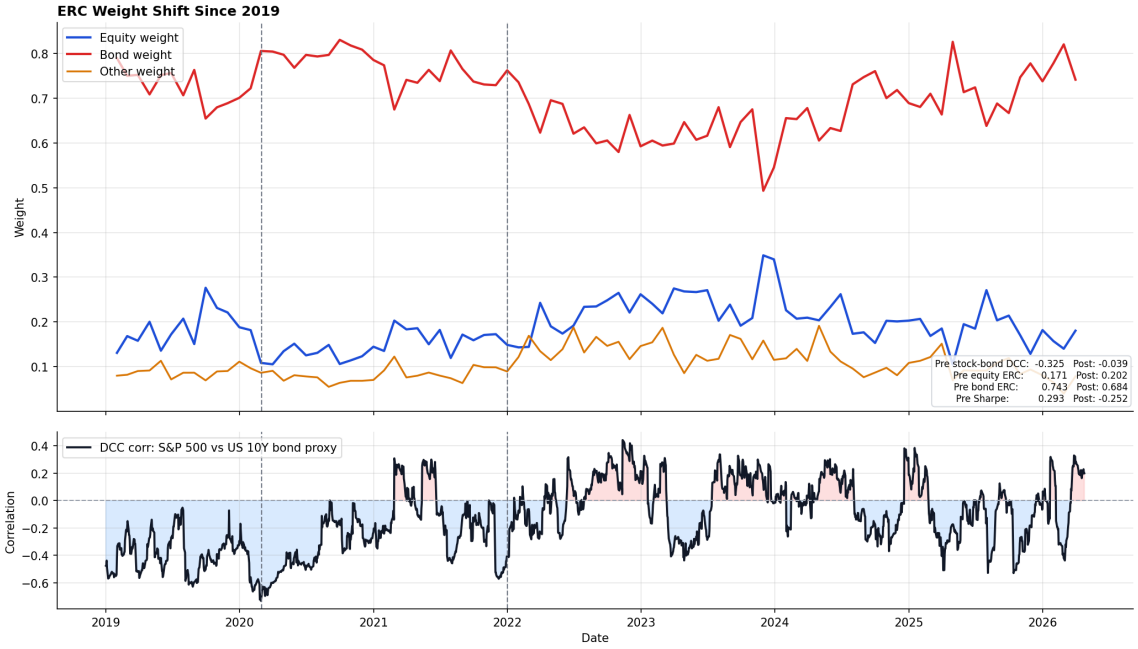


Figure 11: Since 2019, ERC allocations shifted as bond diversification weakened after the inflation and rate-hiking cycle.

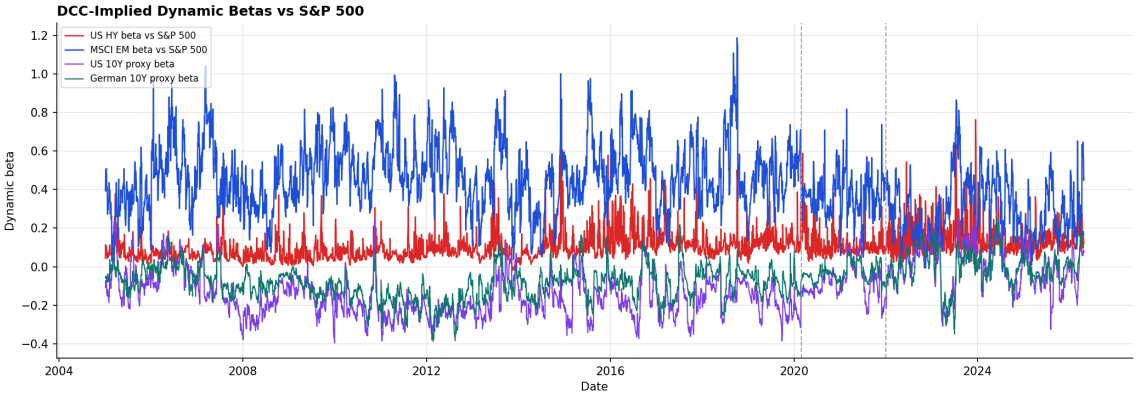


Figure 12: DCC-implied betas show how asset sensitivities to the S&P 500 changed over time.

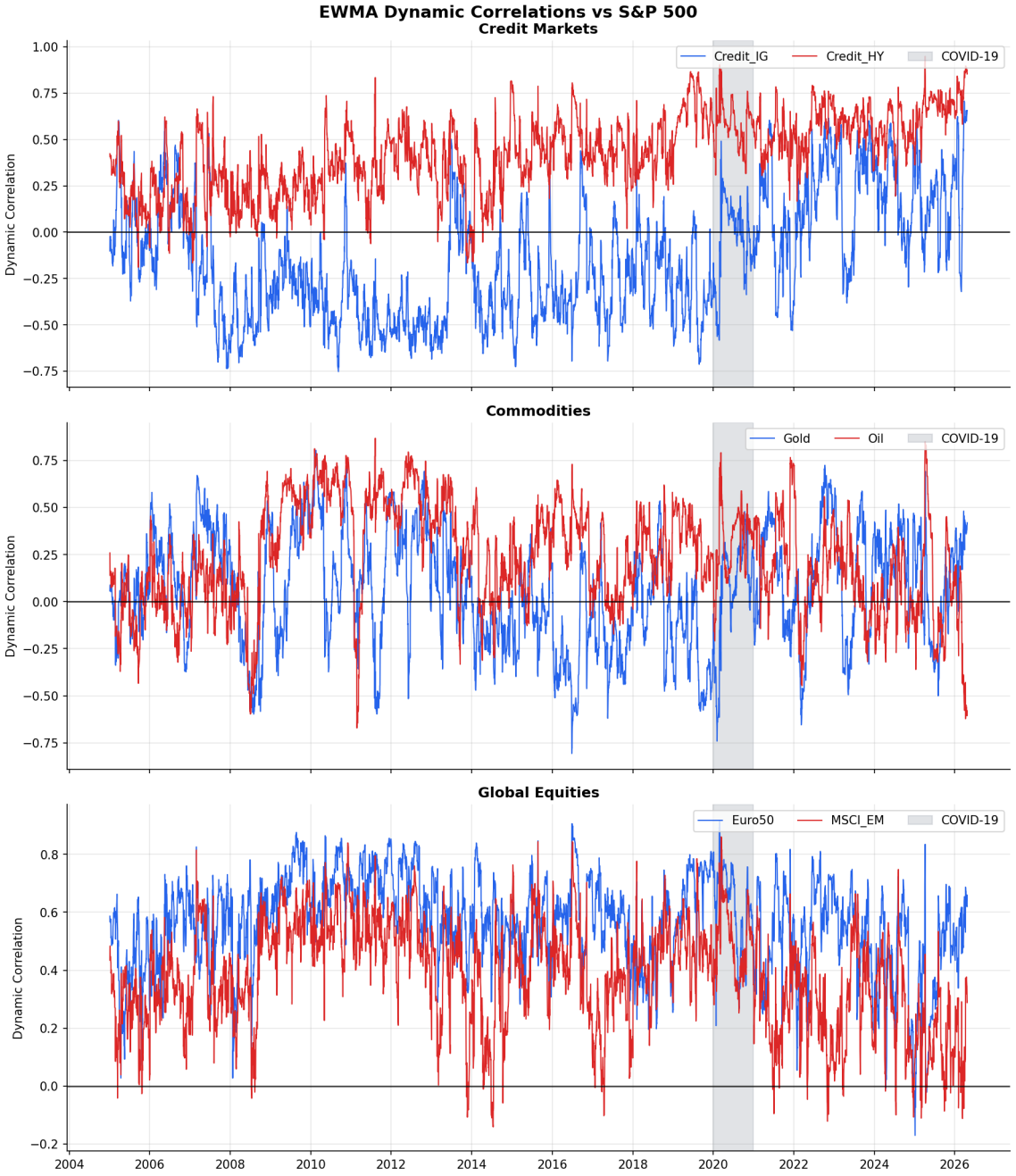


Figure 13: Broad-market dynamic correlations confirm that the post-COVID shift is visible across several asset classes.